

Describe Data by Range and Interquartile Range

Lesson 2-5

Name: Type your name **Date:** Today's date **Class:** Class period



TODAY'S OBJECTIVES

Content Objective: I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

Language Objective: I can explain what the spread of a data set is using the words range, quartile, interquartile range, and vary.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK – FILL EACH BLANK WITH THE BEST WORD

Range and Interquartile Range

Range

Quartile

Interquartile range

Variability

Measure of variation

Extreme value



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The range uses the values; the interquartile range uses the .

2

Subtract the least value from the value to find the range.

3

Numbers that split ordered data into four equal parts are

.

4

The spread of the middle half of the data, $Q3 - Q1$, is the

.

5 How spread out the numbers in a data set are is its .

6 A single number describing how spread out a data set is is a

.

7 The least or greatest value, far from the others, is an

.

Write About the Math

Which class scored better overall, and how do the range and the IQR show it?

¿A qué clase le fue mejor en general, y cómo lo muestran el rango y el rango intercuartílico?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Range and Interquartile Range**

Rango y rango intercuartílico

☐ **Range**
Rango

☐ **Quartile**
Cuartil

☐ **Interquartile range**
Rango intercuartílico

☐ **Variability**
Variabilidad

Model: *Equal medians do not make two data sets the same — you still need to know how spread out the scores are.* — Students name a measure of spread rather than another measure of center.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Class A's range is ____ and Class B's range is ____, so Class A's scores are more ____.** **Class A's IQR is ____ and Class B's IQR is ____, so the middle half of Class B is packed into a ____ stretch of points.** **Because 75% of Class B scored ____ or more, Class B did better ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**

Mi evidencia es _____, entonces _____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.